



Hole Doping Reduces the Coercive Field in Ferroelectric HfO₂

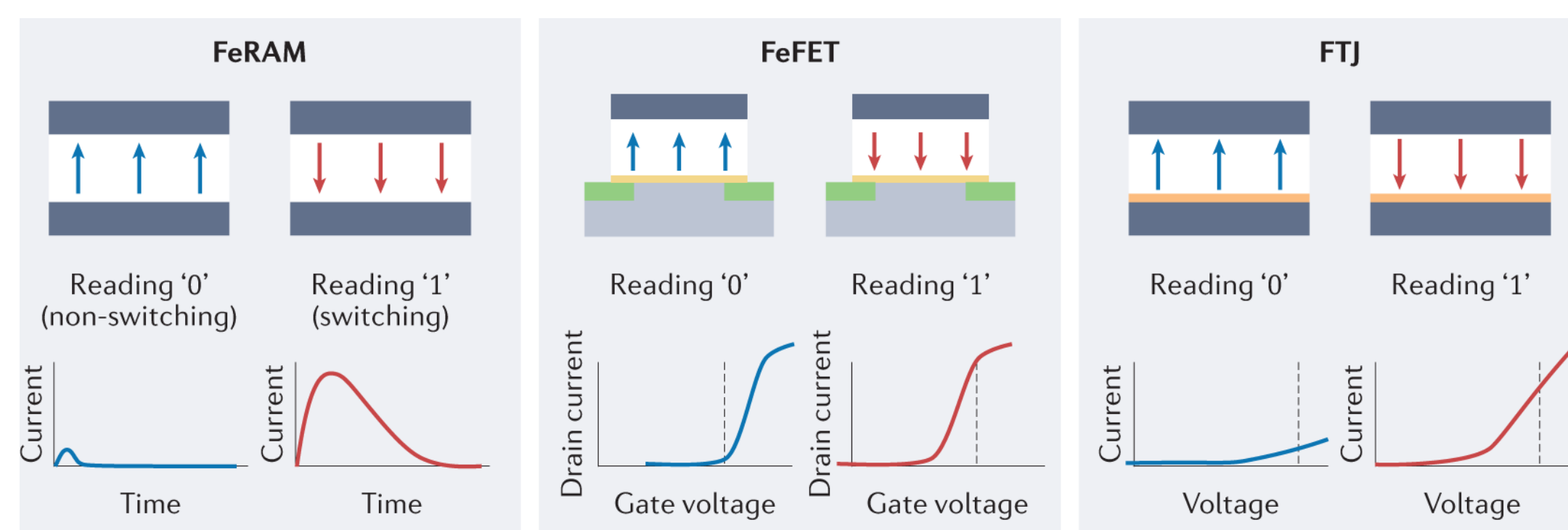
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1. Introduction

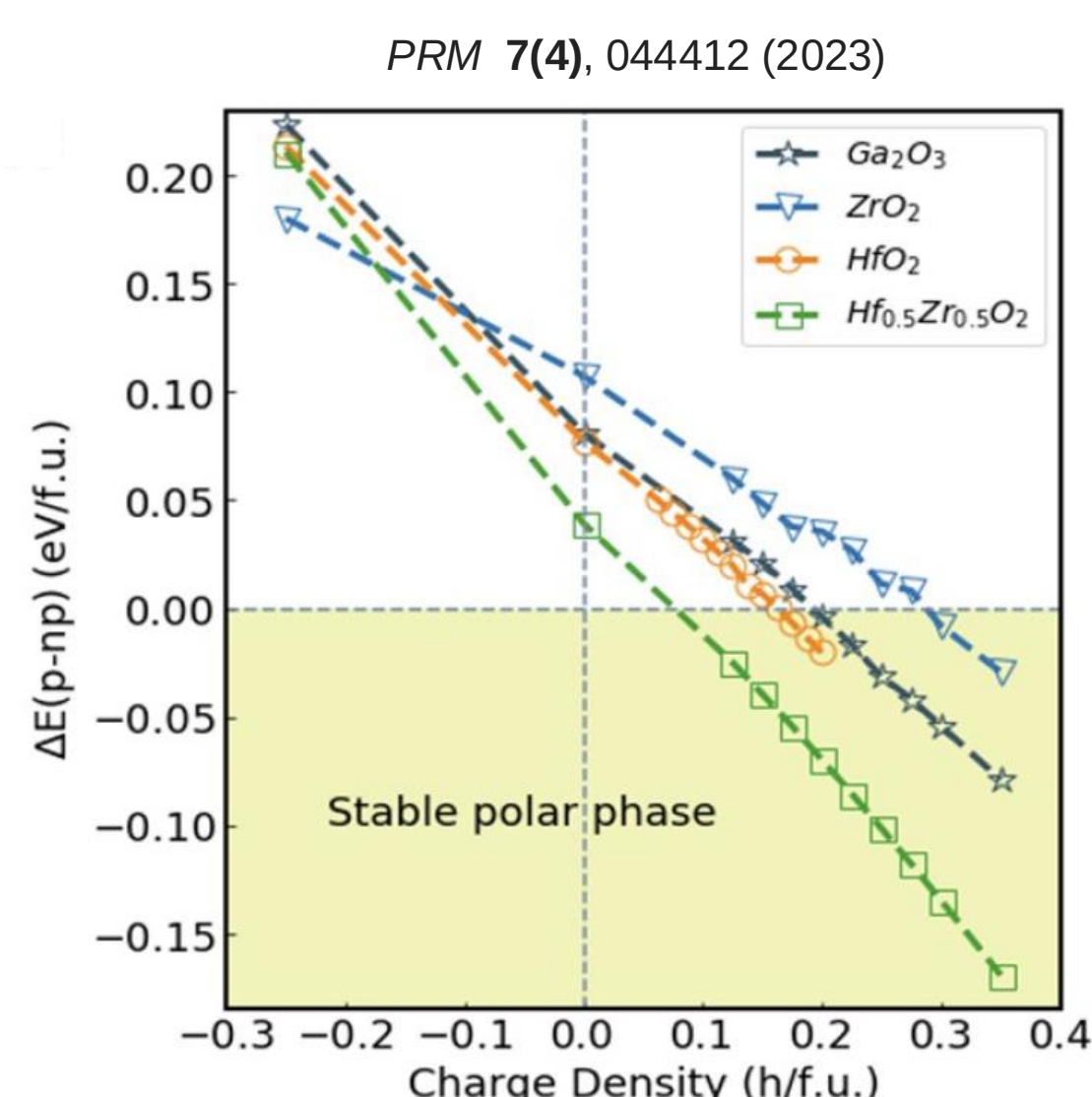
- Ferroelectrics, analogous to ferromagnets, can be used in low-power memory devices.
 - HfO₂ has been integrated with Si-based devices and shows robust ferroelectricity.
- However, HfO₂ has a high coercive field (E_c) which increases the voltage required to use the device.



2. Objective

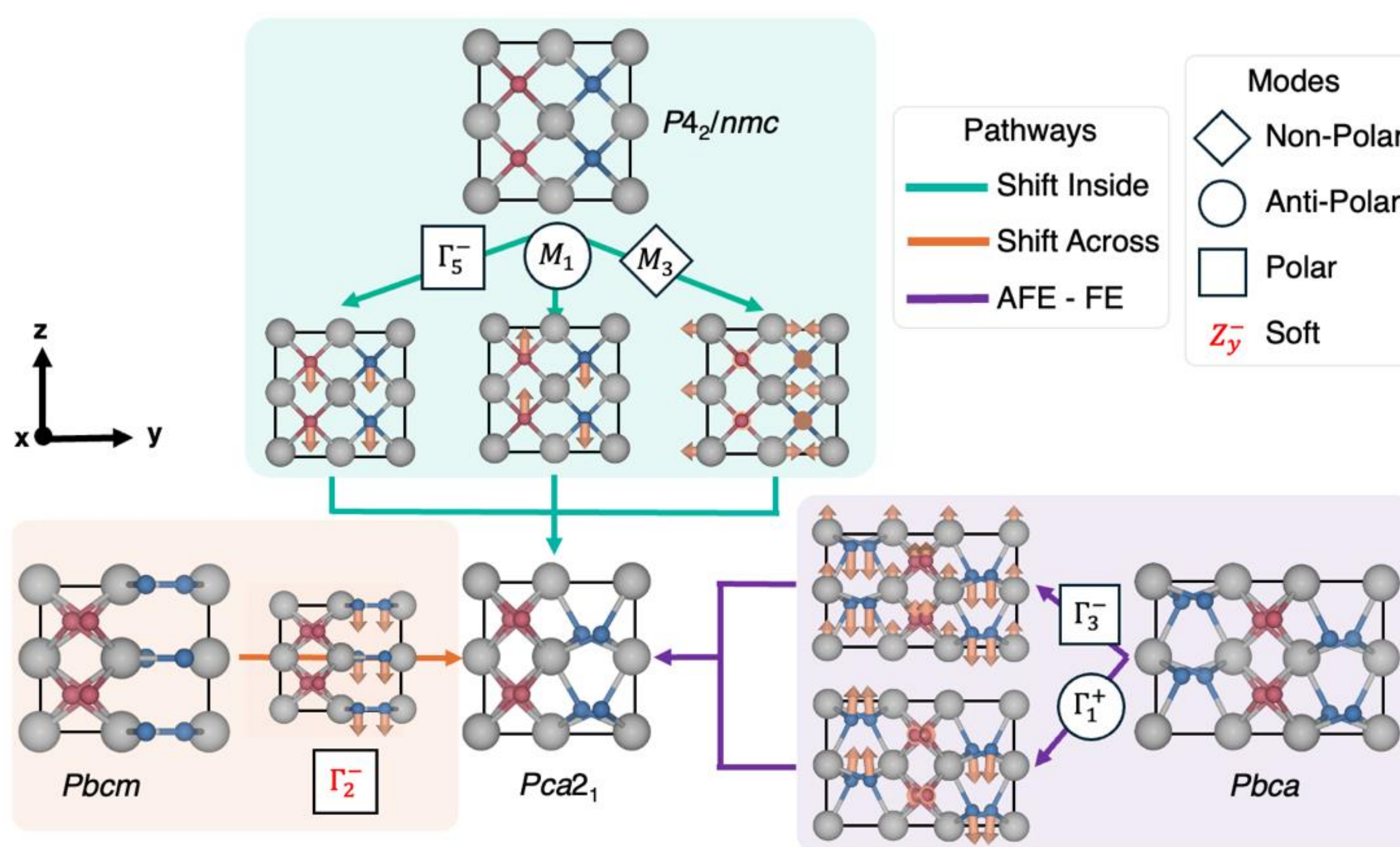
- Can introduction of localized charges reduce the coercive field?

Hole doping stabilizes polar phase in HfO₂

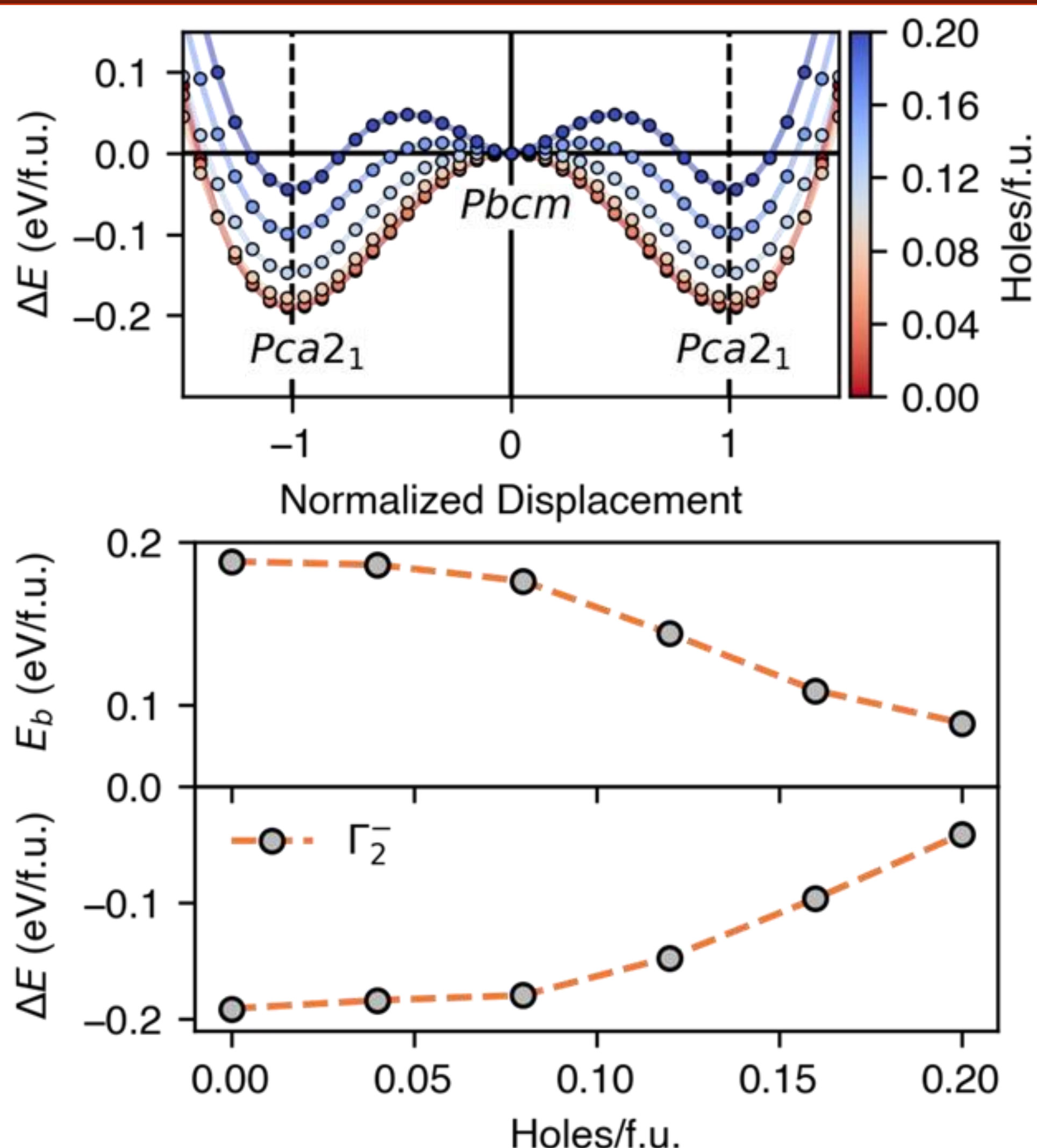


Y, La, Ba – (Hf, Zr)O₂ systems show enhanced ferroelectricity → *Chem Mater* **29**(21): 9102-9109 (2017)

Multiple transition pathways in HfO₂, characterized by different phonon modes

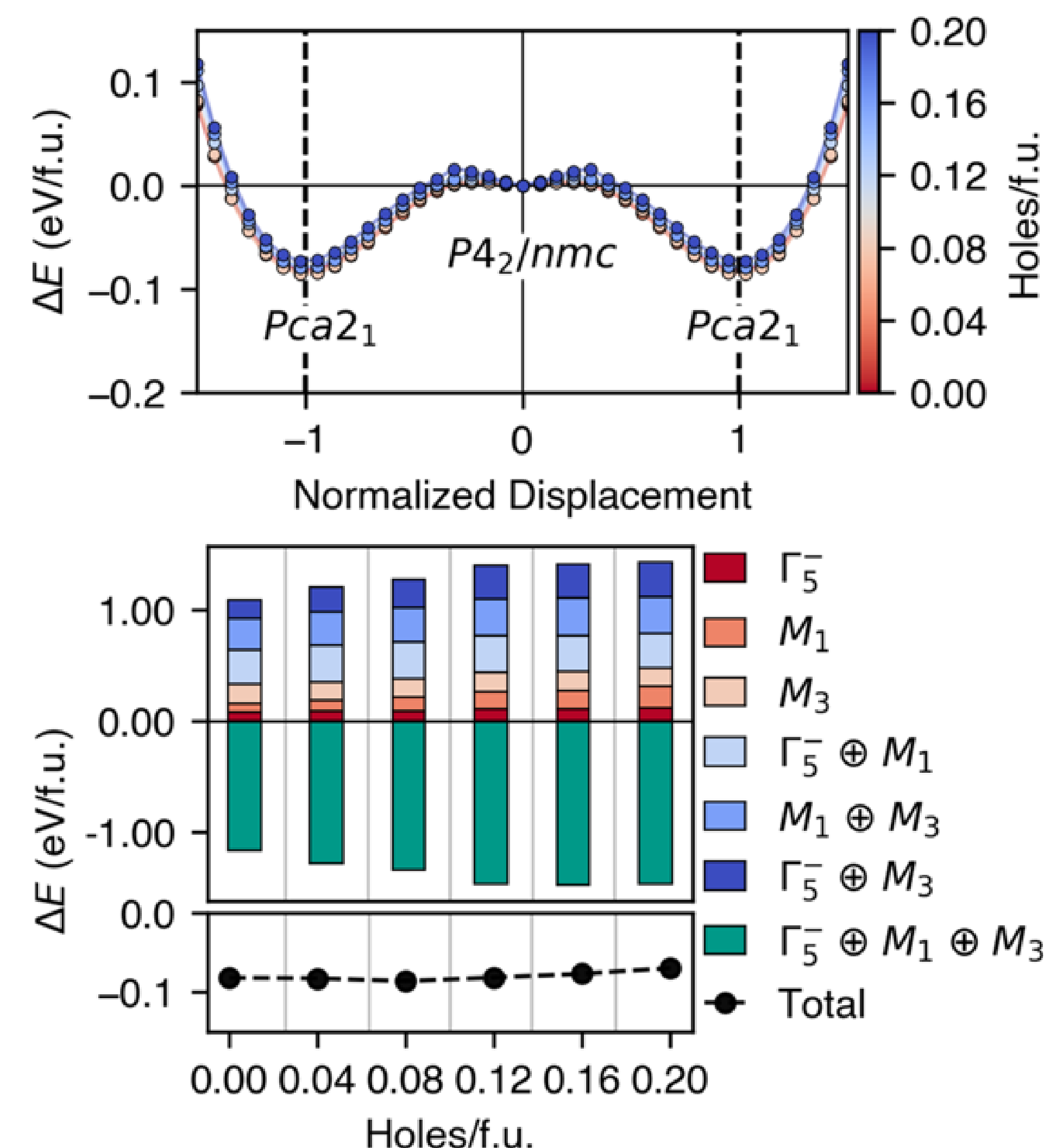


3. How do holes tune the Shift Across Pathway?



- SA pathway barrier reduces considerably.
- Pbcm phase has soft polar mode.
- Soft polar mode decreases with added holes → reducing the barrier energy

4. How do holes couple with modes in Shift Inside Pathway?

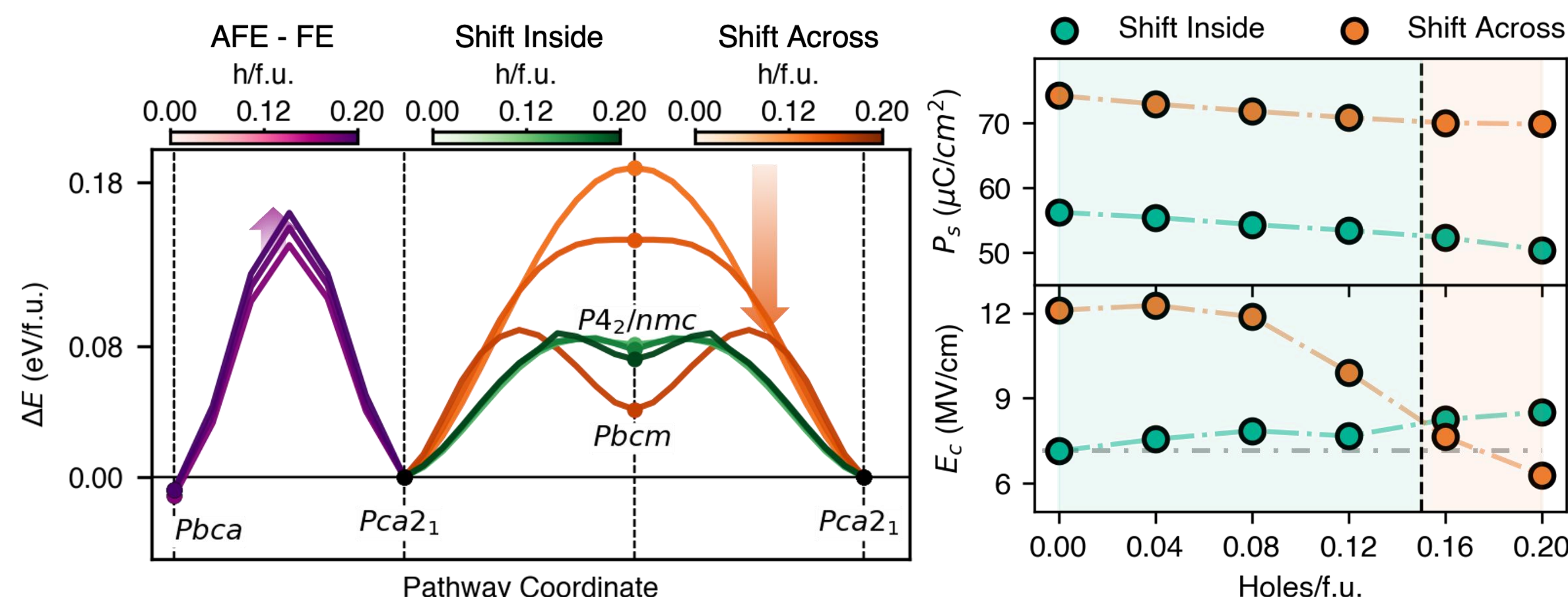


The Shift Inside pathway has negligible change in barrier.

P4₂/nmc has no soft modes; there is polar + antipolar mode coupling

The various couplings of the three modes cancel each other out to keep the barrier constant.

5. Implications on Coercive Field



The AFE-FE pathway barrier increases slightly with hole doping

The coercive field of the Shift Inside Pathway increases slightly and becomes unfavorable at higher doping concentrations

The coercive field of the Shift Across Pathway reduces to 6 MV/cm at 0.2 h/f.u.

6. Conclusions and Future Work

- Doping above 0.2 h/f.u. can stabilize the metastable switching pathway in HfO₂ leading to a 14% reduction in coercive field.

Future directions

- Explore different ways to dope the charges like electrostatic gating and charged domain walls.
- Investigating the coupling between doping and other factors like strain and temperature.

7. Acknowledgement

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